

ADVANCED LINEAR ALGEBRA HOMEWORK 6

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P.4.9. If $A \in \mathbf{M}_n$ is to be represented as $A = XY^\top$ for some $X, Y \in \mathbf{M}_{n \times r}$, explain why r cannot be smaller than $\text{rank}(A)$.

Solution. We claim that the columns of A must lie in the column space of X . To see this we write

$$Y^\top = \begin{bmatrix} y_1^\top \\ y_2^\top \\ \vdots \\ y_r^\top \end{bmatrix},$$

where $y_i \in \mathbb{F}^n$ are columns of Y . Then we have that

$$XY^\top = \sum_{k=1}^r x_k y_k^\top.$$

Now choose an arbitrary vector $v \in \mathbb{F}^n$,

$$Av = XY^\top v = X(Y^\top v)$$

with $Y^\top v \in \mathbb{F}^r$. So Av is a linear combination of the r columns of X , which gives us $\text{col}(A) \subseteq \text{col}(X)$. So

$$\text{rank}(A) = \dim(\text{col}(A)) \leq \dim(\text{col}(X)) = \text{rank}(X) \leq r$$

So $\text{rank}(A) \leq r$, and we are done. □

P.4.10. For $X \in \mathbf{M}_{m \times n}$, $\nu(X)$ denote the nullity of X .

- (a) Show that $\nu(X^\top) = \nu(X) + m - n$.
- (b) Let $A \in \mathbf{M}_{m \times k}$ and $B \in \mathbf{M}_{k \times n}$. Show that the rank inequality given by

$$\text{rank}(A) + \text{rank}(B) - k \leq \text{rank}(AB)$$

is equivalent to

$$(1) \quad \nu(AB) \leq \nu(A) + \nu(B).$$

(c) If $A, B \in \mathbf{M}_n$, show that the rank inequality given by

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

is equivalent to

$$(2) \quad \max\{\nu(A), \nu(B)\} \leq \nu(AB)$$

The inequalities (1) and (2) are known as *Sylvester's law of nullity*.

(d) Consider $A = \begin{bmatrix} 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$. Show that the inequality (2) need not be valid if the matrices are not square.

Proof.

□

P.4.13. Let $\mathcal{V}$ be a nonzero n -dimensional vector space over the field $\mathbb{F}$, and let $T \in \mathfrak{L}(\mathcal{V}, \mathbb{F})$ be a nonzero linear transformation. Why is $\dim(\ker(T)) = n - 1$

Solution.

□

P.4.35. For each of the following matrices, compute its reduced row echelon form, its pivot column decomposition, and bases for its column space and null space.

$$(a) \quad \begin{bmatrix} -1 & -2 & 2 & 3 & 3 \\ 3 & 6 & 1 & 5 & -1 \\ 2 & 4 & -1 & 0 & 2 \end{bmatrix} \quad (b) \quad \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 2 & 2 & 3 & 2 \end{bmatrix}.$$

Solution.

□

P.4.38. Suppose that $A \in \mathbf{M}_{6 \times 4}$ is row equivalent to

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(a) Find a basis for $\text{null}(A)$.

(b) What is $\dim(\text{null}(A^\top))$

Solution.

□